

Transcriptomics teaching

DTU Bioengineering

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Dataset 1: *Staphylococcus aureus*

- **Paper:** <https://doi.org/10.1099/mgen.0.000598>

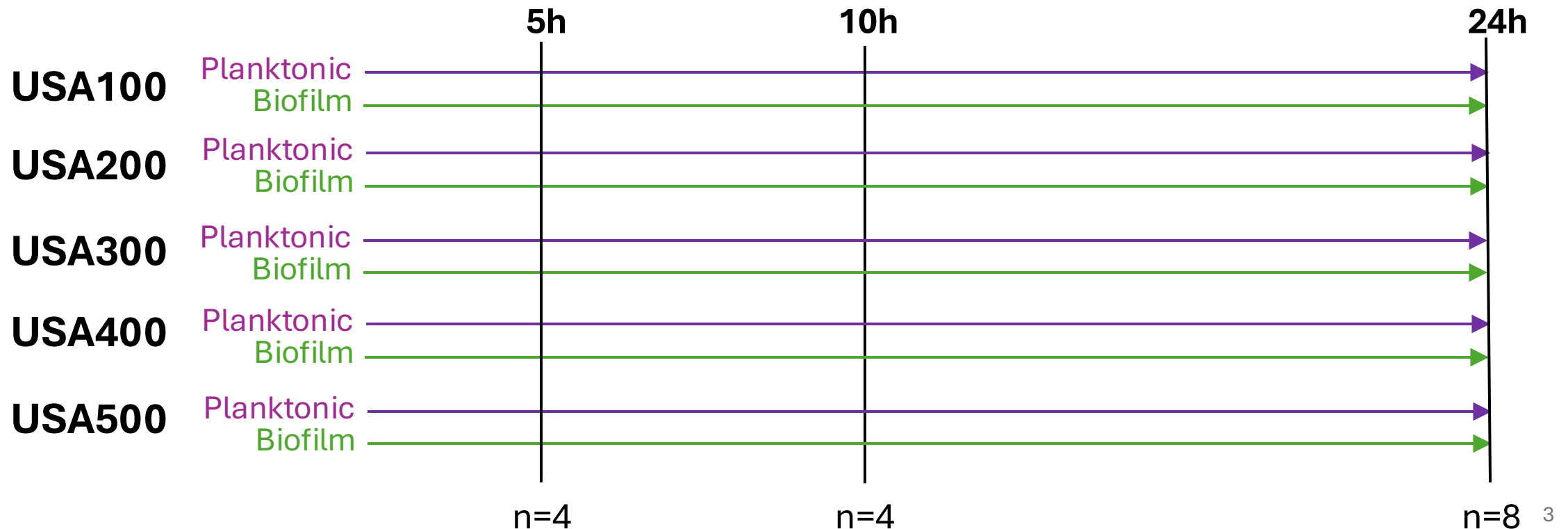
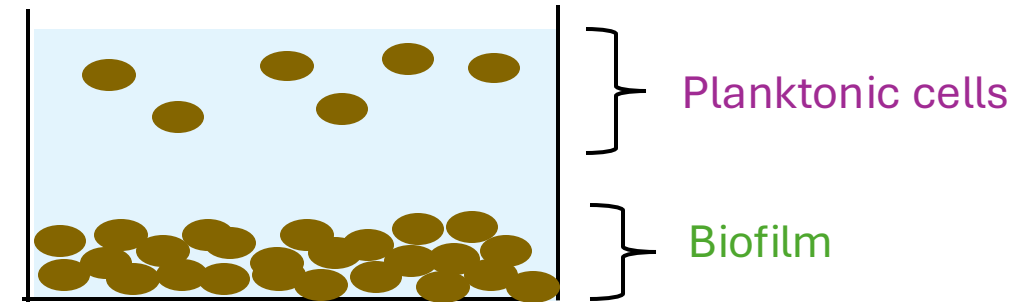
S. aureus strains investigated

Strains	Clonal type*	Origin†	Reference	Sequence type
N315	USA100	HA-MRSA, pharyngeal smear, Japan	(22 , 103)	ST5
MRSA252	USA200	HA-MRSA, postoperative infection, UK	(104)	ST36
LAC	USA300	CA-MRSA, skin/soft tissue, USA	(105)	ST8
MW2	USA400	CA-MRSA, septicemia/septic arthritis, USA	(106)	ST1
NRS385	USA500	HA-MRSA, bloodstream, USA	(22)	ST8
*Clonal type of each strain as determined by PFGE in previous studies.				
†Presentation of infection: HA, nosocomial; CA, community-acquired.				

Dataset 1: *Staphylococcus aureus*

Experimental design:

- Static cultivation of each strain up to 24h
- Time course RNAseq sampling
- Sample size of study: 160
- Sample size per strain: 32 (16 planktonic + 16 biofilm)
- Sample size per time point-strain-lifestyle: see illustration below



Dataset 1: *Staphylococcus aureus*

Biofilm formation

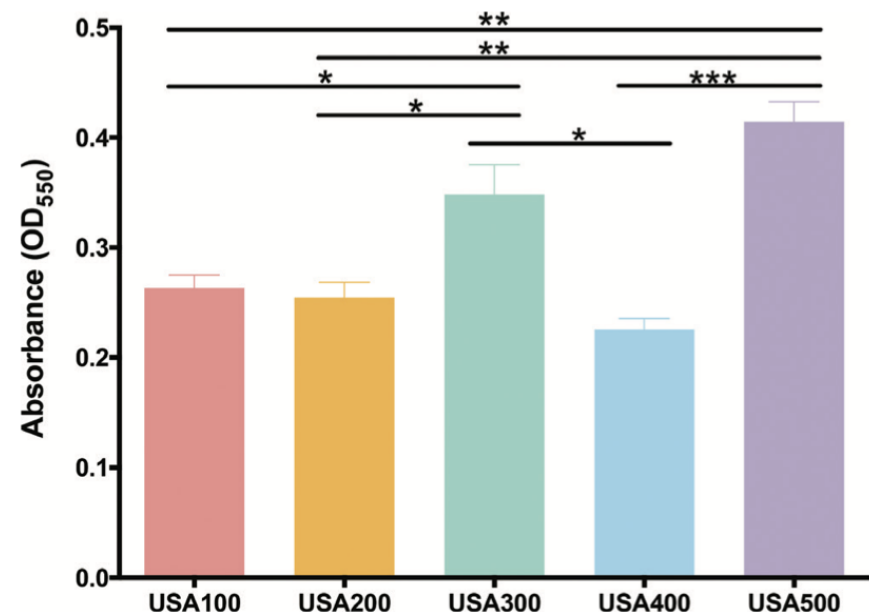


Fig. 2. An assessment of biofilm formation across diverse MRSA clonal lineages. Static biofilms were grown in 96-well plates in TSB for 24 h. Biofilm formation was quantified by crystal-violet staining, followed by 100% ethanol elution and OD₅₅₀ measurement. Assays were performed in biological triplicate with eight technical replicates. Error bars represent $\pm$ SEM and Student's *t*-test was used to determine statistical significance. *, $P < 0.05$; **, $P \leq 0.01$; ***, $P \leq 0.001$.

Differential gene expression analysis

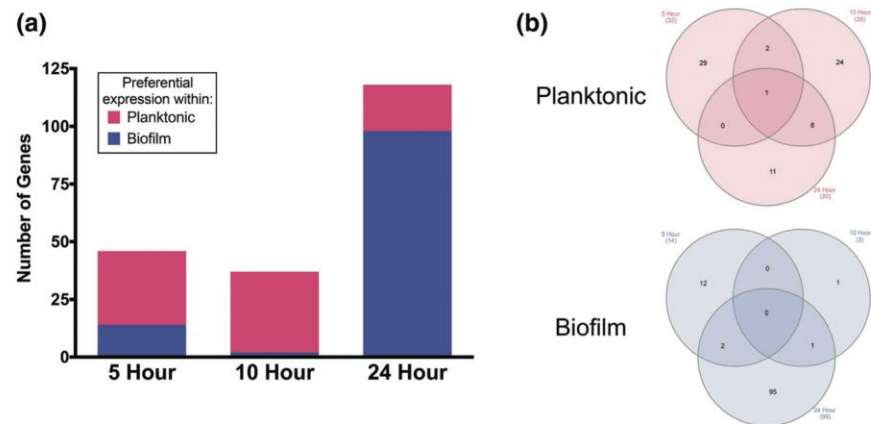


Fig. 5. Differential gene expression occurs primarily in ageing biofilms. (a) The number of genes with >threefold preferential expression in biofilm or planktonic populations conserved across all strains at individual timepoints. (b) Venn diagram comparing biofilm- and planktonic-associated genes from (a) at each timepoint.

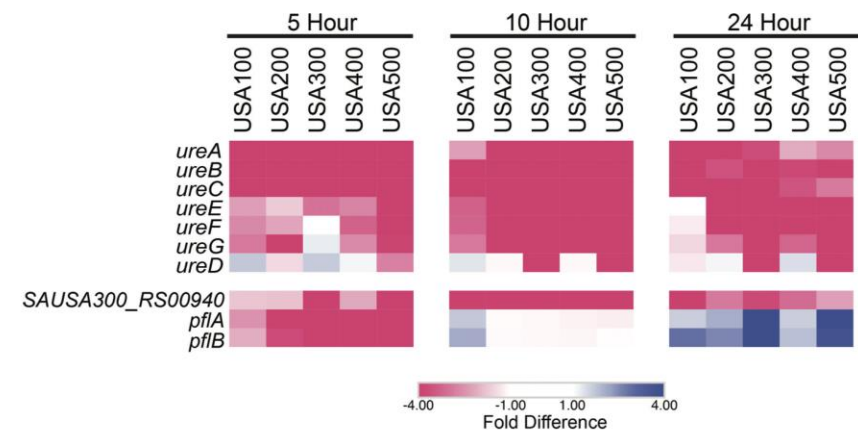


Fig. 6. Biofilms show evidence of a formic acid metabolic response during later stages of growth. Shown are heatmaps depicting preferential gene expression in biofilm (blue) or planktonic (red) populations over time. Values for colours were assigned based on differences in expression between biofilm and planktonic cell population for each timepoint and each strain, all analysed independently.

Dataset 1: *Staphylococcus aureus*

Suggested data analysis

- The paper always does a comparison of biofilm vs planktonic per strain
- **Suggestion 1**
 - PCAs for each, USA100 and USA500
 - From there decide which samples to choose for DEG+GSEA analysis (simple treatment-control design of one time point)
 - Differential gene expression analysis + GSEA
- **Suggestion 2**
 - Perform time-course analysis (but this adds more complexity in the design)

Dataset 1: *Staphylococcus aureus*

gmt files for GSEA

To be looked for

Dataset 2: *Homo sapiens*

- **Paper:** <https://doi.org/10.1371/journal.pone.0099625>
- Supplementary paper associated with the dataset entry: <https://doi.org/10.1038/s41467-022-35739-8>

Background (from abstract)

- Asthma is a chronic inflammatory respiratory disease that affects over 300 million people worldwide
- Glucocorticoids, like dexamethasone, are a mainstay therapy for asthma because they exert anti-inflammatory effects in multiple lung tissues, including the airway smooth muscle (ASM)
- **The mechanism by which glucocorticoids suppress inflammation in ASM remains poorly understood**

Approach (from abstract and methods)

- Primary ASM (airway smooth muscle) cells were isolated from **four** white aborted lung transplant donors with **no chronic illness**
- Primary ASM cells were cultivated and treated with dexamethasone (1 μ M for 18 hours) or vehicle (control).

So far in the paper it is unclear if they

(i) just test DEX on cells from healthy donors or

(ii) if they "mimic asthma" by inducing it in cell line cultivations artificially

Dataset 2: *Homo sapiens*

- In the paper, the authors only dexamethasone (DEX) treatment versus control in RNAseq experiments
- The dataset has additional samples where cells seemed to be treated also with albuterol and dexamethasone+albuterol
- **For the course, I recommend to stick to the dexamethasone (n=4) versus control (n=4) comparison but we may need to account for individual differences of cell donors (see samples below)**

Sample_id	Primary cell line from patient	treatment
N052611_dexamethasone	N052611	Treated: DEX (1microM)
N052611_untreated	N052611	Untreated: vehicle
N061011_dexamethasone	N061011	Treated: DEX (1microM)
N061011_untreated	N061011	Untreated: vehicle
N080611_dexamethasone	N080611	Treated: DEX (1microM)
N080611_untreated	N080611	Untreated: vehicle
N61311_dexamethasone	N61311	Treated: DEX (1microM)
N61311_untreated	N61311	Untreated: vehicle

Dataset 2: *Homo sapiens*

- **For the course, I recommend to stick to the dexamethasone (n=4) versus control (n=4) comparison but we may need to account for individual differences of cell donors (see samples below)**
- The paper does a comparison of DEX_treated_vs_control (I believe considering patient confounding)
- **Suggestion 1**
 - PCA for DEX-treated and untreated samples
 - From here it might be possible already to decide if patient confounding (e.g. due to genetic background) is an issue and needs to be considered for the DESeq2 design formula
 - Differential gene expression analysis + GSEA
 - It could be informative to perform DEG analysis with and without accounting for inter-patient variability (for pure teaching purposes as suggested by Lasse)

Data processing summary

Organism	Strain	Accession	n samples	Link to Seqera run	Path in Azure (Seqera production)	Ref	Comments
<i>S. aureus</i>	usa100	PRJNA685119 GSE163153	32	https://cloud.seqera.io/orgs/DTU-Biosustain/workspaces/seqera-production/watch/3mpCtQMcnlLhx4	workstoragedls/seqera/results/teaching_bioengineering_08_Oct_2026/usa100_all_samples_01Sept_E16_replicates/	https://doi.org/10.1099/mgen.0.000598	
	usa500	PRJNA685119 GSE163153	32	https://cloud.seqera.io/orgs/DTU-Biosustain/workspaces/seqera-production/watch/3PKNfROWiwGHxP	workstoragedls/seqera/results/teaching_bioengineering_08_Oct_2026/usa500_all_samples_01Sept_E16_replicates/	https://doi.org/10.1099/mgen.0.000598	Ref genome not available but usa300 is believed to be the closest relative → aligned against usa300 ref genome
<i>H. sapiens</i>	-	PRJNA229998 GSE52778	16	https://cloud.seqera.io/orgs/DTU-Biosustain/workspaces/seqera-production/watch/4kA0drjz5CKbPl	workstoragedls/seqera/results/teaching_bioengineering_08_Oct_2026/hsapiens_PRJNA229998_GSE52778_3Sept_E16_v326_all_samples/	https://doi.org/10.1371/journal.pone.0099625 https://doi.org/10.1038/s41467-022-35739-8	Aligned against hg19 (GRCh37) as was done in the paper